

Training and Fine-Tuning Strategy

Zero-Shot Prompting

Currently, the application relies on prompt engineering and zero-shot model instruction parameters. The Gemini model uses system context turns to adopt specific agent personas without weights modifications. A clean Prompt will help to generate a good answer for the user but still this has some vulnerabilities which will be corrected while exploring SNAP

Future Fine-Tuning Roadmap

For target deployments, fine-tuning a small model (e.g., Gemma 2B or Gemma 9B) could yield structural compliance equal to larger models while reducing hosting costs.

1. Curate Dataset: Collect 5,000 high-quality query-to-structured-JSON pairs.
2. Execute PEFT (LoRA): Run Low-Rank Adaptation training.
3. Quantize Model: Run 4-bit/8-bit quantization for browser-side WebGPU acceleration.